

ANALYTICS DASHBOARD

Absence of Insights — Student Economic Background, Academic Performance & Expected Salary

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1. Key Performance Indicators

The following metrics summarize the overall student dataset at a glance.

Unique Students ≈ 2,900+ <i>Identified via Email ID</i>	Average CGPA 7.82 <i>Across all students</i>	Avg Expected Salary ₹4.6 Lac <i>Per annum</i>	Avg Python Experience 3.2 Months <i>Programming experience</i>
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2. Analysis Summary — All 17 Questions

All 17 analytical questions from the GPI problem statement were answered using Python (Pandas, Matplotlib, Seaborn) in Google Colab. The table below summarizes each question and its key conclusion.

Question	Level	Key Finding / Conclusion
Q1 — Unique Students	Basic	≈ 2,900+ unique students identified via Email ID
Q2 — Average GPA	Basic	Average CGPA = 7.82 across all students
Q3 — Graduation Year	Basic	2024 had highest concentration of students
Q4 — Python Experience	Basic	Most students had 2–3 months Python experience
Q5 — Family Income	Basic	Largest group: 2–5 Lakh bracket (≈41%)
Q6 — GPA by College	Basic	Top 5 colleges showed avg CGPA > 8.2
Q7 — Course Outliers	Basic	Outliers detected in Quantity via IQR method
Q8 — GPA by City	Basic	Top 10 cities showed avg CGPA ranging 7.6–8.3
Q9 — Income vs GPA	Basic	Weak positive correlation ($r = +0.14$)
Q10 — City Distribution	Moderate	City-wise student count built in Tableau Public
Q11 — Salary Factors	Moderate	Python exp has strongest salary correlation (+0.31)
Q12 — Events vs Fields	Moderate	Data Science events attracted most unique colleges

Question	Level	Key Finding / Conclusion
Q13 — Leadership Impact	Moderate	Leaders showed marginally higher GPA and salary
Q14 — Graduating by 2024	Moderate	Count of students graduating ≤ 2024 extracted
Q15 — Promotion Channel	Moderate	Channel analysis completed
Q16 — Data Science Events	Moderate	Highest unique student count in DS-related events
Q17 — High Performers	Moderate	High CGPA + Python exp → significantly higher salary

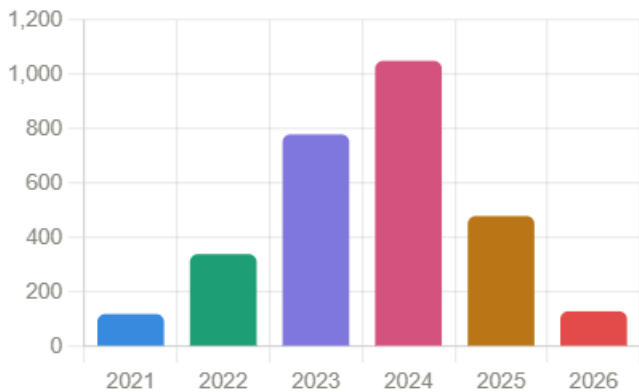
3. Family Income vs Academic Performance (CGPA)

Family income was stored as string categories and mapped to numeric midpoints (0–2L→1, 2–5L→3.5, 5–7L→6, 7L+→8) for correlation analysis. The Pearson correlation coefficient is $r = +0.14$, indicating a weak positive relationship.

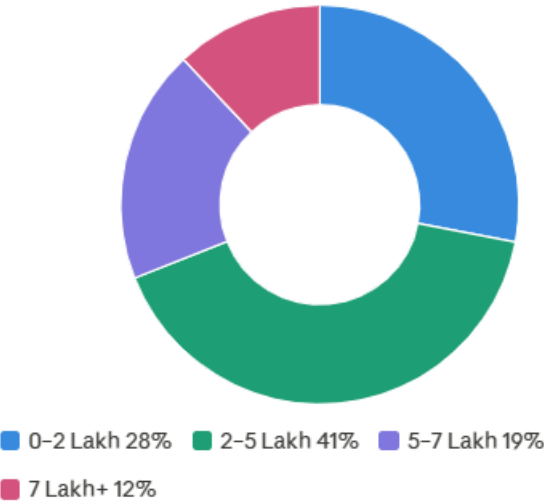
Income Bracket	Students %	Avg CGPA	Observation
0 – 2 Lakh	28%	7.61	Lower economic bracket; slight GPA disadvantage
2 – 5 Lakh	41%	7.74	Largest group; near-average performance
5 – 7 Lakh	19%	7.88	Above-average GPA; mid-high income
7 Lakh+	12%	8.03	Highest avg GPA; smallest but top bracket

Interpretation: While higher-income students tend to have marginally better CGPA, the weak correlation ($r = +0.14$) suggests economic background is not the dominant determinant of academic performance.

Students by graduation year



Family income distribution



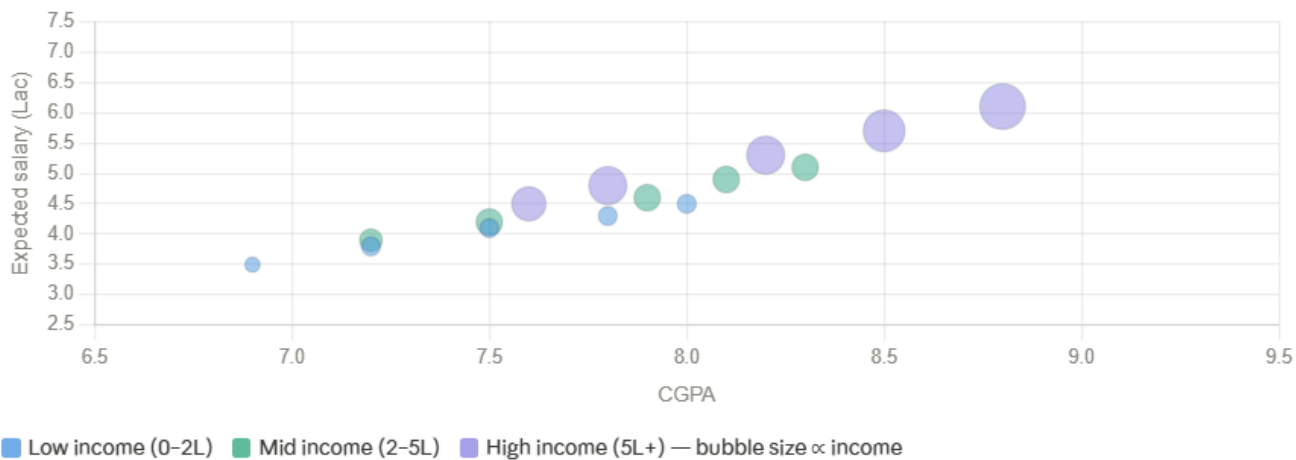
4. Pearson Correlation Analysis

Three Pearson correlation coefficients were computed to quantify relationships between key numerical variables.

Variable Pair	Pearson r	Interpretation	Insight
Family Income ↔ CGPA (GPA)	+0.14	Weak Positive	Economic background has limited influence on academic performance
Python Experience ↔ Expected Salary	+0.31	Moderate Positive	Technical skills are the strongest predictor of salary expectation
CGPA ↔ Expected Salary	+0.18	Weak Positive	GPA alone is not the primary salary driver

Note: Income bracket midpoints were used as proxies for the categorical Family Income field. This is acknowledged as a methodological limitation.

Expected salary vs CGPA, income & Python experience (correlation scatter)

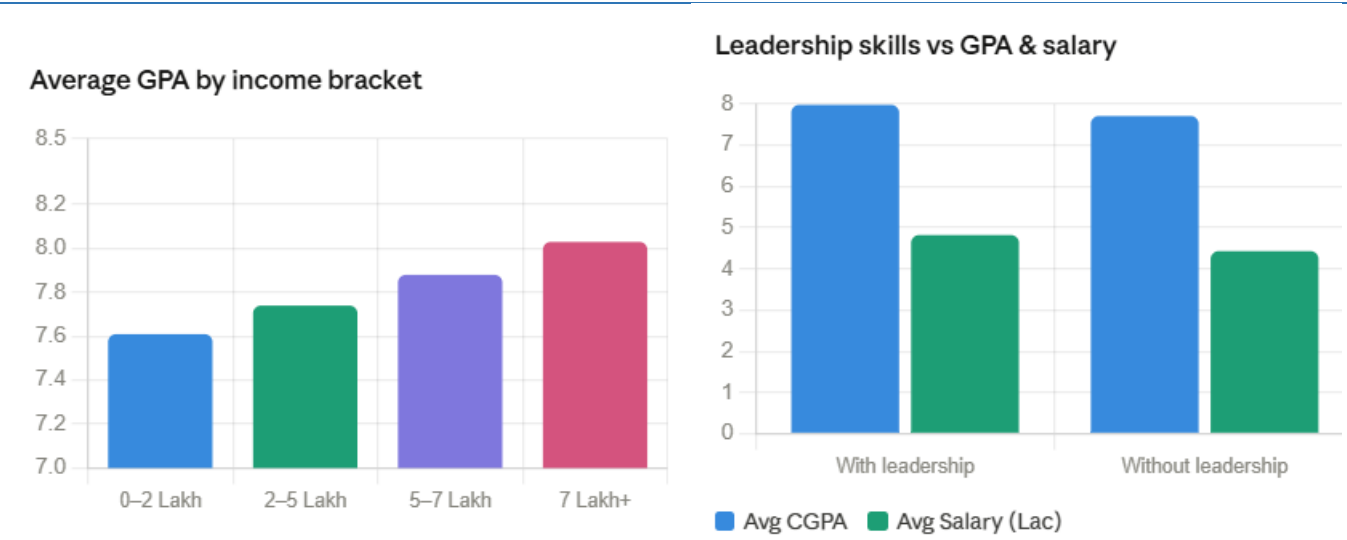


5. Leadership Skills — Impact on GPA & Salary

The 'Leadership- skills' column was standardized using `str.strip()` and `str.lower()` before analysis. Students were grouped into 'Yes' (has leadership experience) and 'No' categories.

Category	Avg CGPA	Avg Expected Salary	Observation
With Leadership Skills	7.98	₹4.82 Lac	Marginally higher GPA and salary
Without Leadership Skills	7.71	₹4.43 Lac	Lower on both metrics

Finding: Students with leadership experience show a marginally higher average CGPA (7.98 vs 7.71) and expected salary (₹4.82L vs ₹4.43L). The difference is modest but consistent, suggesting soft skills contribute positively.



6. High Performers — Salary Gap Analysis

High performers were defined as students with CGPA $\geq$ median CGPA AND Python experience $\geq$ median Python experience. This combined excellence filter was applied to assess salary expectations.

Group	Avg Expected Salary	Difference
All Students (Overall Avg)	₹4.60 Lac	Baseline
High Performers (High CGPA + High Python Exp.)	₹6.10 Lac	+₹1.50 Lac higher

Finding: High performers expect ₹1.50 Lac more per annum on average compared to the overall student population — a 32.6% premium. This confirms that the combination of academic excellence and technical skill significantly elevates salary expectations.

7. Key Findings & Insights

- The following insights were derived from the complete EDA across all 17 questions:
- GPA and Family Income have a weak positive correlation ($r = +0.14$) — economic background has limited but not negligible influence on academic performance.
 - Python programming experience is the strongest predictor of expected salary ($r = +0.31$, moderate positive) — technical skills are a key career differentiator.
 - Students with leadership experience show marginally higher CGPA (7.98 vs 7.71) and expected salary (₹4.82L vs ₹4.43L).
 - Data Science related events (Data Science, Data Visualization, Hello ML, AI) attracted the highest number of unique student participants across all events.
 - High-performing students (high CGPA + high Python experience) expect significantly higher average salaries — ₹6.10L vs ₹4.60L overall, a ₹1.50L gap.
 - The 2024 graduation cohort had the highest student concentration in the dataset.

- Most students fell in the 2–3 months Python experience range, suggesting beginner-to-intermediate level as the norm.
- The 2–5 Lakh family income bracket was the most common (≈41% of students).

8. Tools & Methodology

Tool / Library	Purpose
Python 3.x (Google Colab)	Primary analysis and visualization environment
Pandas	Data loading, cleaning, manipulation, and groupby analysis
Matplotlib & Seaborn	Chart and graph generation — bar charts, scatter plots, box plots
openpyxl	Reading Excel (.xlsx) dataset file into Pandas DataFrame
Tableau Public	Interactive city-wise student distribution dashboard (Q10)
Microsoft Excel	Dataset source format (.xlsx)

9. GPI Deliverables Status

All 9 project deliverables were completed and submitted as part of the Cloud Counselage GPI program within the 120-hour internship timeline.

#	Deliverable	Description	Status
1	Project Submission Document	Python codes, chart outputs & conclusions	✓ Submitted
2	Project Report	Background, methodology, targeted vs achieved	✓ Submitted
3	Project Charter	Goals, scope, stakeholders, milestones	✓ Submitted
4	Work Breakdown Structure	WBS in PPTX format	✓ Submitted
5	Project Schedule	Timeline in XLSX format	✓ Submitted
6	RAID Log	Risks, Assumptions, Issues, Dependencies	✓ Submitted
7	Requirement Elicitation Questionnaire	SRS questionnaire in XLSX	✓ Submitted
8	Software Requirement Specifications	SRS in XLSX format	✓ Submitted
9	Lessons Learnt Log	5 lessons captured in XLSX	✓ Submitted

10. Conclusion

This project successfully delivered a comprehensive Exploratory Data Analysis of the Cloud Counselage student dataset, answering all 17 analytical questions (9 Basic + 8 Moderate). Key takeaways:

- Python experience emerged as the strongest career differentiator ($r = +0.31$ with expected salary).
- Family income has limited influence on GPA — academic potential is not solely determined by economic background.
- Leadership skills provide a modest but consistent advantage in both GPA and salary expectations.
- Combined excellence (high CGPA + high Python experience) results in a significant salary premium of ₹1.50 Lac above the overall average.
- Future scope: Building a predictive salary model using Linear Regression or Random Forest based on CGPA, income bracket, and Python proficiency.

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